



PAPER ID-310805

Printed Page: 1 of 2
Subject Code: RCAI502

Roll No:

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MCA-INT
(SEM V) THEORY EXAMINATION 2024-25
PROGRAMMING WITH MATLAB

TIME: 3 HRS**M.MARKS: 70**

Note: Attempt all Sections. In case of any missing data; choose suitably.

SECTION A**1. Attempt all questions in brief.****2 x 07 = 14**

Q no.	Question
a.	Write any two features of MATLAB.
b.	Differentiate between uint8 and double datatype.
c.	Define the term string with example.
d.	Why numel() and ndims() functions are used?
e.	What do you mean by function handle?
f.	What is mesh grid?
g.	Briefly explain functions to solve linear equation.

SECTION B**2. Attempt any three of the following:****07 x 3 = 21**

a.	Discuss various mathematical functions with proper example.
b.	Explain any five functions used to change the shape of a two-dimensional matrix.
c.	Write a program to perform addition and multiplication of two matrix using nested loops.
d.	Discuss any three specialized three-dimensional plots.
e.	Explain steps and write a program to find roots of a polynomial.

SECTION C**3. Attempt any one part of the following:****07 x 1 = 07**

a.	What is a variable? Explain various types of variables with example.
b.	Explain various application areas of MATLAB.

4. Attempt any one part of the following:**07 x 1 = 07**

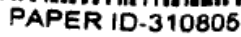
a.	Create a 4X4 matrix and do the following – (i) Count the number of elements of the matrix (ii) Perform sum of all diagonal elements of the matrix (iii) Convert the matrix into one-dimensional array (iv) Transpose the matrix (v) Rotate the matrix by 90 degrees
b.	Discuss different ways to compare two strings with proper example.

5. Attempt any one part of the following:**07 x 1 = 07**

a.	What do you mean by a decision control statement? Explain any two decision control statements used in MATLAB with example.
b.	What is a function? Write a program to demonstrate the concept of function and sub-function.

6. Attempt any one part of the following:**07 x 1 = 07**

a.	Why do we create subplots? Illustrate the concept of subplot with the help of a program (plot 4 subplots).
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| b. | What is a two-dimensional plot? Discuss various functions used to plot bar chart in MATLAB. |
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7. Attempt any one part of the following:

07 x 1 = 07

- | | |
|--|---|
| 7. Attempt any one part of the following. | |
| a. | Explain any five functions used for data analysis. |
| b. | Write short notes on –
(i) Eigen Value and Eigen Vector
(ii) Functions to compute integrals |

QP25DP7_005

QP25DP1_000

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